Function Transformations: Shifts, Scalings, & Roots

1. Summary of Elementary Transformations

Let $y = f(x)$ be a given parent function and $c > 0$ a positive constant:

- **Vertical Translations:**
  - $y = f(x) + c$: Shifts graph **up** by c units.
  - $y = f(x) - c$: Shifts graph **down** by c units.
- **Horizontal Translations:**
  - $y = f(x - c)$: Shifts graph **right** by c units (replaces x with $x - c$).
  - $y = f(x + c)$: Shifts graph **left** by c units (replaces x with $x + c$).
- **Scalings (Stretches and Compressions):**
  - $y = c \cdot f(x)$ $(c > 1)$: Vertical stretch by factor c.
  - $y = f(cx)$ $(c > 1)$: **Horizontal compression** by factor c (inputs are divided by c).

2. Classroom Polynomial Case Study: Tracking Roots

Consider the cubic polynomial with given roots at $x = -1, 1, 3$:

$$f(x) = (x + 1)(x - 1)(x - 3)$$

Task: Construct the transformed function $g(x) = f\big(2(x - 1)\big)$, simplify its factors, and determine its new roots.

- **Step 1: Substitute the transformed input $2(x - 1)$ into each factor:**

$$\begin{aligned} g(x) &= \Big(\big[2(x-1)\big] + 1\Big)\Big(\big[2(x-1)\big] - 1\Big)\Big(\big[2(x-1)\big] - 3\Big) \\ &= \big(2x - 2 + 1\big)\big(2x - 2 - 1\big)\big(2x - 2 - 3\big) \\ &= (\mathbf{2x - 1})(\mathbf{2x - 3})(\mathbf{2x - 5}) \end{aligned}$$

- **Step 2: Solve for the new roots ($g(x) = 0$):**
  - $2x - 1 = 0 \implies \mathbf{x = \frac{1}{2}}$
  - $2x - 3 = 0 \implies \mathbf{x = \frac{3}{2}}$
  - $2x - 5 = 0 \implies \mathbf{x = \frac{5}{2}}$

Geometric Transformation Verification:
The transformation $f\big(2(x - 1)\big)$ applies two horizontal operations to the original x-coordinates:

1. **Horizontal Compression by 2:** Divides each original root by 2:

$$-1 \to -0.5, \quad 1 \to 0.5, \quad 3 \to 1.5$$

2. **Horizontal Shift Right by 1:** Adds $+1$ to each coordinate:

$$-0.5 + 1 = \mathbf{0.5}, \quad 0.5 + 1 = \mathbf{1.5}, \quad 1.5 + 1 = \mathbf{2.5} \qquad \checkmark$$

Algebra and geometry agree completely!